Pathfinder Software Requirements Specification (SRS)

Overview

Project Name: Pathfinder
Team Members:

- Patrick King - Project Manager
- Maddie McDowell - Front-End and UI Design
- Nic Ortiz - Front-End
- Thomas Powell - Back-End

Brief Description:
Pathfinder is an all-in-one career development web application designed to guide users through every stage of their professional and educational journey. It offers AI-powered career assistance through resume analysis, automated job matching, and networking resources. The platform leverages OpenAI for natural language processing, Firebase for data management, and modern web technologies for a seamless user experience.

Project Scope

This document details the software requirements for Pathfinder, a web application developed using React, HTML, CSS, and Node.js. The target users include high school and college students, job seekers, and professionals seeking career transitions. The primary goal of Pathfinder is to provide users with a streamlined career development experience by integrating multiple tools into a single platform. Features include AI-assisted career guidance, interactive resume-building tools, and seamless job search capabilities.

Component Breakdown

Front-End

- Landing Page: Login/Sign-Up functionality with responsive design for various devices.
- Dashboard: Displays personalized user information and provides access to core features.
- Chatbot UI: AI chatbot accessible on all pages for quick assistance.
- Profile Setup Page: Allows users to enter and edit personal and career preference details.

Back-End

- Authentication & User Management: Handles login and registration using Firebase.
- Career Guidance Engine: Processes user data to provide career recommendations.
- Resume & Job Matching System: Integrates AI-powered resume analysis and automated job matching.
- API Integration Layer: Connects with external services for job listings and learning resources.

AI-Powered Chatbot

- Query Processing: Routes user queries to OpenAI API for response generation.
- Context Management: Retains session context for a conversational experience.

Database & Data Storage

- Firebase Database: Stores user profiles, preferences, and historical data securely.

Deployment

- Hosted on cloud platforms (e.g., AWS or Google Cloud) for scalability and reliability.

Required States and Modes

- **Landing Page State:** Initial page for logging in or signing up.
- **Dashboard State:** Centralized hub displaying user-specific tools and information.
- **Settings State:** Allows customization of account settings and preferences.

Functional Requirements

- FR1: The website shall support user authentication via Firebase.
- FR2: Users shall be able to create and manage career preference profiles.
- FR3: Users shall be able to reset their password via email link.
- FR4: The system shall allow users to upload a profile picture.
- FR5: Users shall be able to update their account settings.
- FR6: The system shall allow deletion of user accounts.

- FR7: The system shall notify users of login from a new device.
- FR8: Users shall be able to manage notification preferences.
- FR9: Users shall receive confirmation emails after registration.
- FR10: Users shall be able to enable two-factor authentication.
- FR11: The website shall provide personalized career suggestions based on a user questionnaire.
- FR12: The system shall match jobs based on user-selected career paths.
- FR13: The job-matching system shall refresh results every 24 hours.
- FR14: The job-matching system shall allow users to filter results based on salary, location, and remote work availability.
- FR15: The system shall recommend upskilling courses based on job listings that do not fully match user qualifications.
- FR16: Users shall be able to retake the questionnaire at any time.
- FR17: Users shall be able to save or bookmark job suggestions.
- FR18: Users shall be able to rate the relevance of job suggestions.
- FR19: Users shall receive alerts when new matching jobs are found.
- FR20: Users shall be able to hide or remove unwanted job listings.
- FR21: The website shall analyze uploaded resumes and suggest improvements.
- FR22: The resume tool shall suggest edits based on job descriptions.
- FR23: The system shall generate tailored cover letter templates.
- FR24: Users shall be able to download resume/cover letter drafts as PDFs.
- FR25: The AI chatbot shall provide career advice through a floating interface.

External Interface Requirements

- EIR1: Integrate with job board APIs for fetching job listings.
- EIR2: Use Firebase for secure data storage and authentication.
- EIR3: Connect to LinkedIn and Coursera APIs for learning and networking resources.

Internal Data Requirements

- IDR1: Store user profiles and preferences securely in Firebase.
- IDR2: Store career questionnaire results and resume feedback data.
- IDR3: Log chatbot interactions for session continuity.

Adaptation Requirements

- AR1: Adapt layout responsively for mobile, tablet, and desktop.

- AR2: Support both portrait and landscape orientation on mobile.

Security Requirements

- SecR1: Encrypt user data during transmission and storage.
- SecR2: Restrict access to personalized features to authenticated users.

Environmental Requirements

- ER1: Support modern browsers including Chrome, Firefox, Safari, and Edge.

Computer Resource Requirements

- CRR1: Run efficiently on low-spec devices with no extra plugins required.

Software Quality Factors

- SQF1: Provide intuitive UI to reduce need for instructions.
- SQF2: Notify users in advance of system downtime.

Design & Implementation Constraints

- DIC1: Platform limited to technologies React, Node.js, Firebase, and OpenAI.

Training-related Requirements

- TRR1: Include a step-by-step tutorial for new users.
- TRR2: Provide contextual tooltips across the app.

Other Requirements

- OR1: Support accessibility with screen readers and keyboard nav.
- OR2: Offer high-contrast mode and text resizing options.